Introduction to Full Stack Development

Day 1 - Python Full Stack Course
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1. What is Full Stack Development?

Full Stack Development means building **both sides** of a web application: the part users see and interact with (**frontend**) and the part that works behind the scenes (**backend** and **database**). A **full stack developer** can design the interface, write the server logic, store data, and deploy the finished application.

The word *stack* refers to the set of technologies layered together to build one application.

2. Frontend vs Backend vs Database

Layer	What it does	Examples
Frontend	Everything the user sees and clicks: pages, forms, buttons, layout, animations. Runs in the browser.	HTML, CSS, JavaScript, React.js
Backend	Business logic, authentication, validation, and APIs. Receives requests, processes them, sends responses.	Python, Django, Django REST Framework
Database	Stores and retrieves data permanently, such as users, products, and orders.	SQL, SQLite, MySQL, PostgreSQL

Example: On an online shopping site, the product page you see is the frontend, the code that checks your login and calculates the cart total is the backend, and the saved product list and your order history live in the database.

3. Client-Server Architecture

Web applications follow the **client-server model**:

- **Client:** the browser or app that sends requests (for example, a user opening a page).
- **Server:** the machine running the backend that receives requests, processes them, and replies.
- **Database:** the server talks to it to read or save data.
- Communication happens over **HTTP/HTTPS**, commonly using **JSON** as the data format.

Basic Architecture Diagram

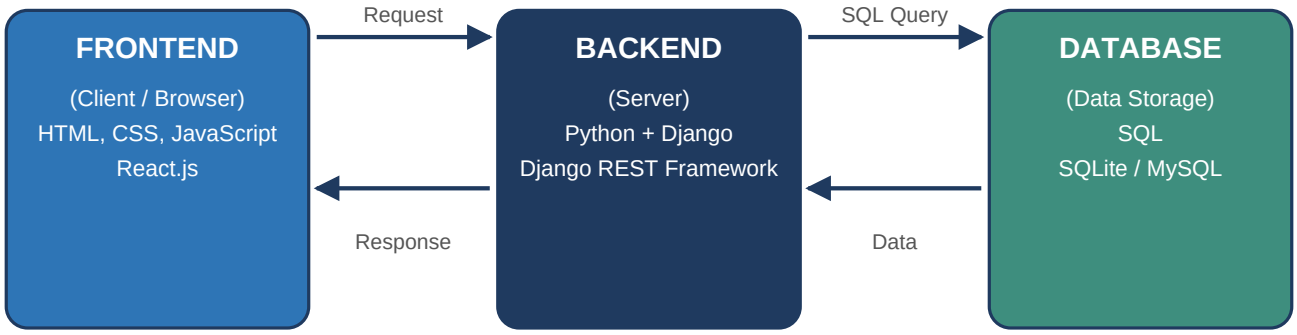

Figure 1: Frontend, backend and database communication.

4. How a Web Application Works

When you use a web app, this sequence happens in a fraction of a second:

1. The user opens a page or clicks a button in the browser (**frontend**).
2. The frontend sends an **HTTP request** (GET, POST, PUT, DELETE) to the server's API.
3. The **backend** receives the request, checks authentication, and applies business logic.
4. The backend runs a **SQL query** to read or write data in the **database**.
5. The database returns the result to the backend.
6. The backend sends a **response** (usually JSON) back to the frontend.
7. The frontend updates the screen to show the result to the user.

HTTP Method	Purpose	Example
GET	Read data	Fetch the list of products
POST	Create data	Register a new user
PUT / PATCH	Update data	Edit profile details
DELETE	Remove data	Delete a task

5. Introduction to Python Full Stack Development

Python Full Stack Development uses **Python** for the backend together with standard web technologies for the frontend. Python is popular because its syntax is simple and readable, it has a huge library ecosystem, and frameworks like Django help build secure, scalable applications quickly.

Typical stack: **React.js (frontend) + Django REST Framework (backend API) + SQL database**.

6. Technologies Overview

Technology	Role	Short description
HTML	Frontend	Defines the structure and content of a web page (headings, paragraphs, forms, links).
CSS	Frontend	Styles the page: colors, fonts, spacing, layout, and responsive design.
JavaScript	Frontend	Adds interactivity: handling clicks, validating forms, calling APIs.
React.js	Frontend	JavaScript library for building reusable UI components and single-page applications.
Python	Backend	General-purpose programming language used to write server-side logic.
Django	Backend	High-level Python web framework with built-in ORM, authentication, and admin panel.
Django REST Framework	Backend	Toolkit on top of Django for building REST APIs that return JSON.
SQL	Database	Language for creating, querying, and managing data in relational databases.

7. Technologies Used in a Python Full Stack Application

Layer	Technologies
Frontend	HTML, CSS, JavaScript, React.js
Backend	Python, Django, Django REST Framework
Database	SQL with SQLite, MySQL or PostgreSQL
API format	REST APIs with JSON
Version control	Git and GitHub
Testing tools	Postman (API testing), browser DevTools
Deployment	Render, Vercel, or similar cloud platforms

8. Summary

- Full stack development covers the frontend, backend, and database of an application.
- The frontend runs in the browser, the backend processes requests, and the database stores data.
- Web apps follow the client-server model, communicating through HTTP requests and responses.
- A Python full stack typically uses HTML, CSS, JavaScript, React.js, Python, Django REST Framework, and SQL.